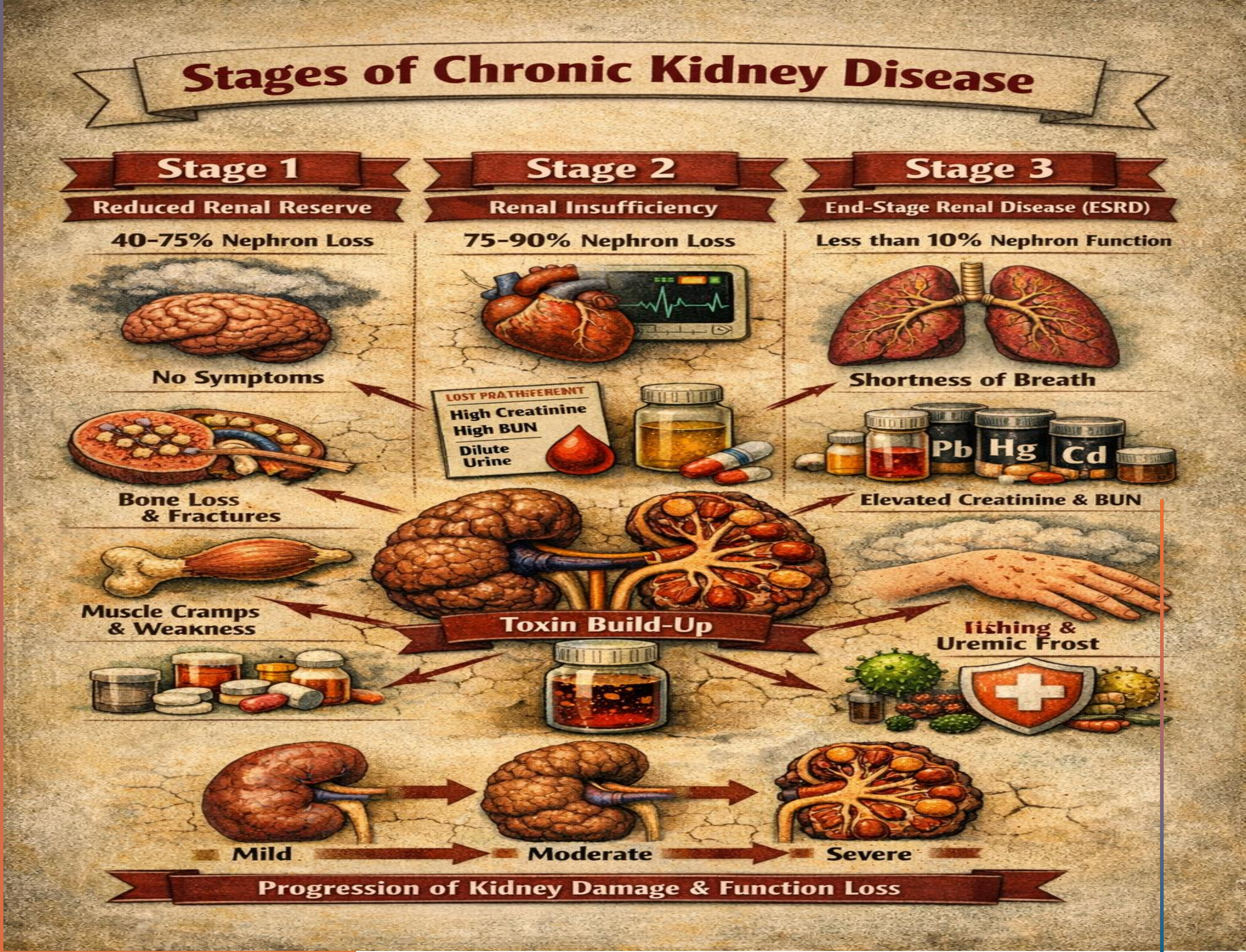

Stages of Chronic Kidney Disease
Stage 1
Reduced Renal Reserve
40–75% Nephron Loss
Stage 2
Renal Insufficiency
75–90% Nephron Loss
Stage 3
End-Stage Renal Disease (ESRD)
Less than 10% Nephron Function
No Symptoms
High Creatinine
High BUN
Dilute Urine
Shortness of Breath
Pb Hg Cd
Bone Loss & Fractures
Elevated Creatinine & BUN
Muscle Cramps & Weakness
Toxin Build-Up
Uremic Frost
Mild
Moderate
Severe
Progression of Kidney Damage & Function Loss

Clinical Manifestations

Cardiovascular manifestations

Hypertension (due to sodium and water retention), heart failure and pulmonary edema (due to fluid overload are among the cardiovascular problems manifested in ESRD.

Dermatologic symptoms

Severe itching (pruritus) is common. The deposit of urea crystals on the skin, is uncommon today because of early and aggressive treatment of ESRD with dialysis.

Other Systemic Manifestations

GI signs and symptoms are common and include anorexia, nausea, and vomiting. Neurologic changes, including altered levels of consciousness, inability to concentrate, muscle twitching, and seizures, have been observed.

Neurologic

Weakness and fatigue; inability to concentrate; disorientation; tremors; seizures; restlessness of legs; burning of soles of feet; behavior changes

<u>Integumentary</u>

Gray-bronze skin color; dry, skin; pruritis; purpura; thin, brittle nails; and thinning hair.

Cardiovascular

Hypertension; pitting edema (feet, hands); per orbital edema; pericardial friction rub; engorged neck veins; pericarditis; hyperkalemia; hyperlipidemia

Pulmonary

Crackles; thick, tenacious sputum; depressed cough reflex; shortness of breath; tachypnea; uremic pneumonitis; “uremic lung”

Gastrointestinal

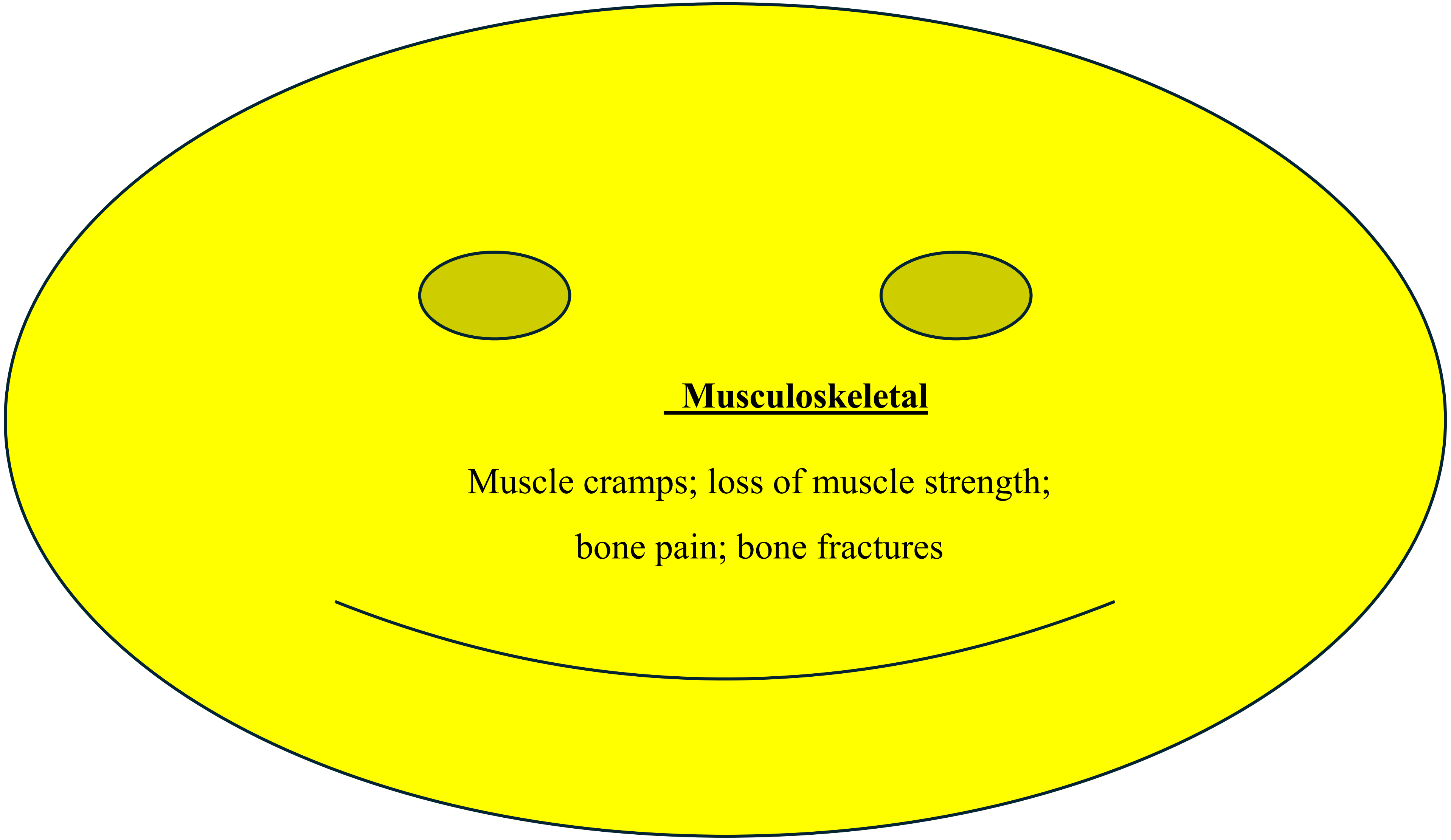
Ammonia odor to breath; metallic taste; mouth ulcerations and bleeding; anorexia, nausea, and vomiting; constipation or diarrhea; bleeding from gastrointestinal tract

Hematologic

Anemia; thrombocytopenia

Reproductive

Amenorrhea; testicular atrophy; infertility


Musculoskeletal
Muscle cramps; loss of muscle strength;
bone pain; bone fractures

Assessment and Diagnostic Findings

Decreased GFR can be detected by obtaining a 24-hour urinalysis for creatinine clearance.

Some patients retain sodium and water, increasing the risk for edema, heart failure, and hypertension

With advanced renal disease, metabolic acidosis occurs because the kidney cannot excrete increased loads of acid.

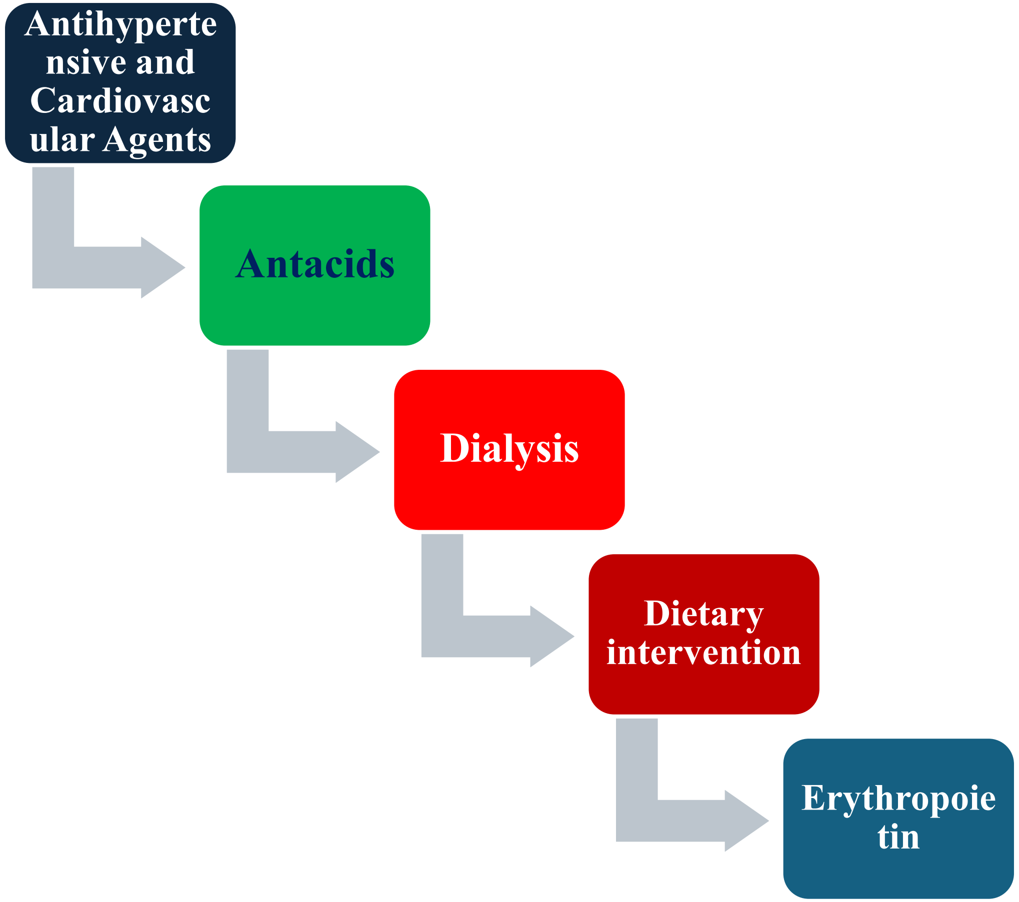
Anemia develops as a result of inadequate erythropoietin production, nutritional deficiencies.

Another major abnormality seen in chronic renal failure is a disorder in calcium and phosphorus metabolism.

Complications

- Pericarditis
- Hyperkalemia
- Anemia
- Hypertension
- Bone disease and metastatic calcifications

Pharmacologic Therapy



create unique image about Nursing Management Examples of potential nursing diagnoses for these patients include the following:

- Excess fluid volume related to decreased urine output, and retention of sodium and water.
- Imbalanced nutrition: less than body requirements related to anorexia, nausea and vomiting

- Deficient knowledge regarding condition and treatment regimen.
- Activity intolerance related to fatigue, anaemia and dialysis procedure.
- Low self-esteem related to dependency.

Dialysis is used to remove fluid and uremic waste products from the body when the kidneys cannot do so.

Acute dialysis is indicated when there is a high level of serum potassium, fluid overload, or increasing acidosis, and severe confusion. It may also be used to remove certain medications or other toxins (poisoning or medication overdose) from the blood.

Chronic or maintenance dialysis is indicated in chronic renal failure, known as end-stage renal disease (ESRD)

Hemodialysis is the most commonly used method of dialysis. It is used for patients who are acutely ill and require short-term dialysis (days to weeks) and for patients with ESRD who require long-term or permanent therapy.

Complications of Hemodialysis

1. Atherosclerotic cardiovascular disease.
2. Disturbances of lipid metabolism
3. Heart failure, and angina pain.
4. Anemia and fatigue
5. Increased dialyzer clotting may occur

6. Gastric ulcers and other gastrointestinal problems.

7. Renal osteodystrophy that produces bone pain and fractures.

8. Heart failure, malnutrition, infection, neuropathy, and pruritis.

9. Up to 85% of people undergoing hemodialysis experience major sleep problems

10. Hypotension may occur during the treatment

11. Painful muscle cramping may occur.

12. Dysrhythmia

13. Chest pain may occur in patients with anemia

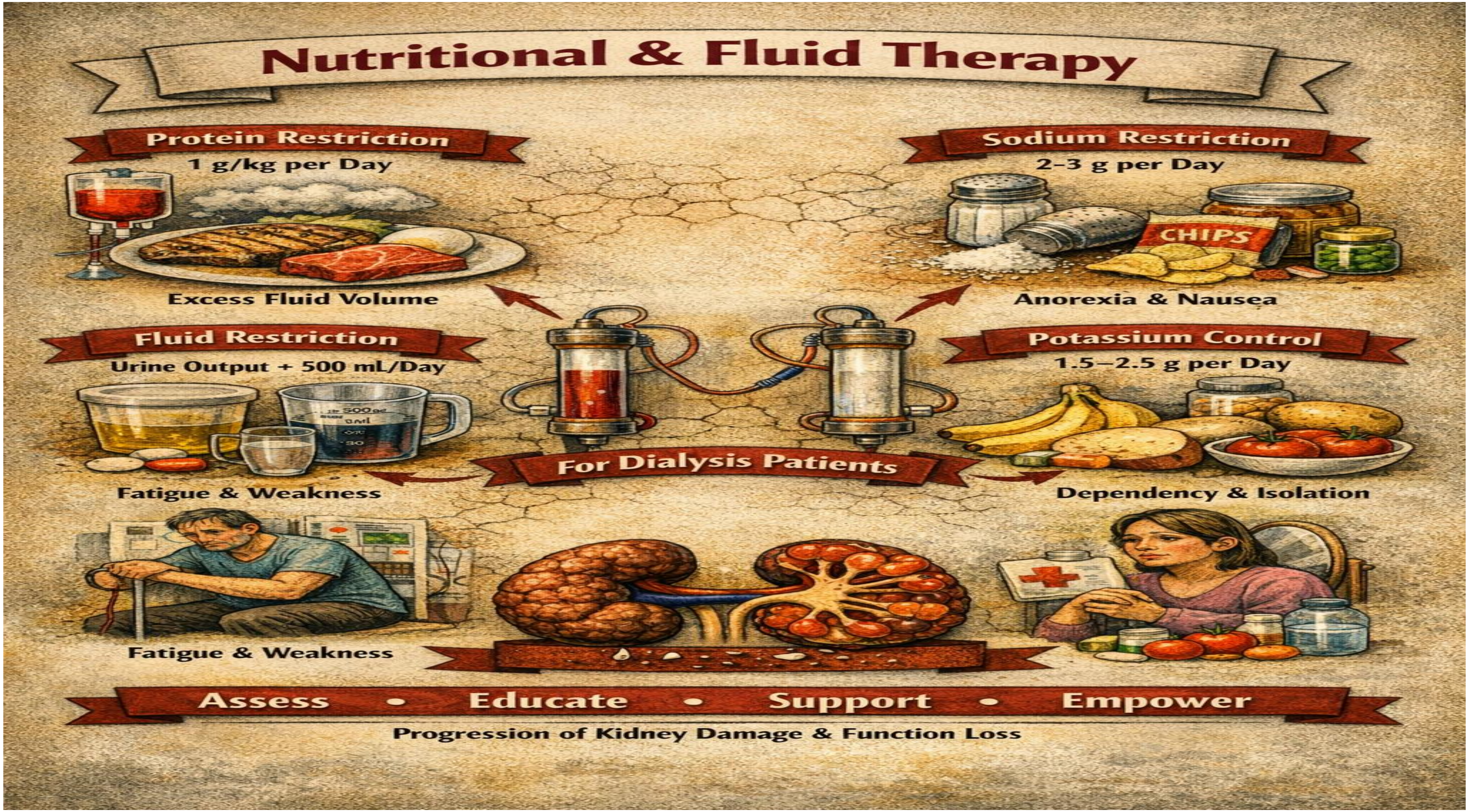
14. Acute allergic reaction and uremic pruritis

Long-Term Management

During dialysis, the patient, the dialyzer, and the dialysate bath require constant monitoring because numerous complications are possible. The nurse in the dialysis unit has an important role in monitoring, supporting, assessing, and educating the patient.

Pharmacological Therapy

Patients undergoing hemodialysis who require medications (e.g., antibiotic agents, antiarrhythmic medications, antihypertensive agents) are monitored closely to ensure that blood and tissue levels of these medications are maintained without toxic accumulation.

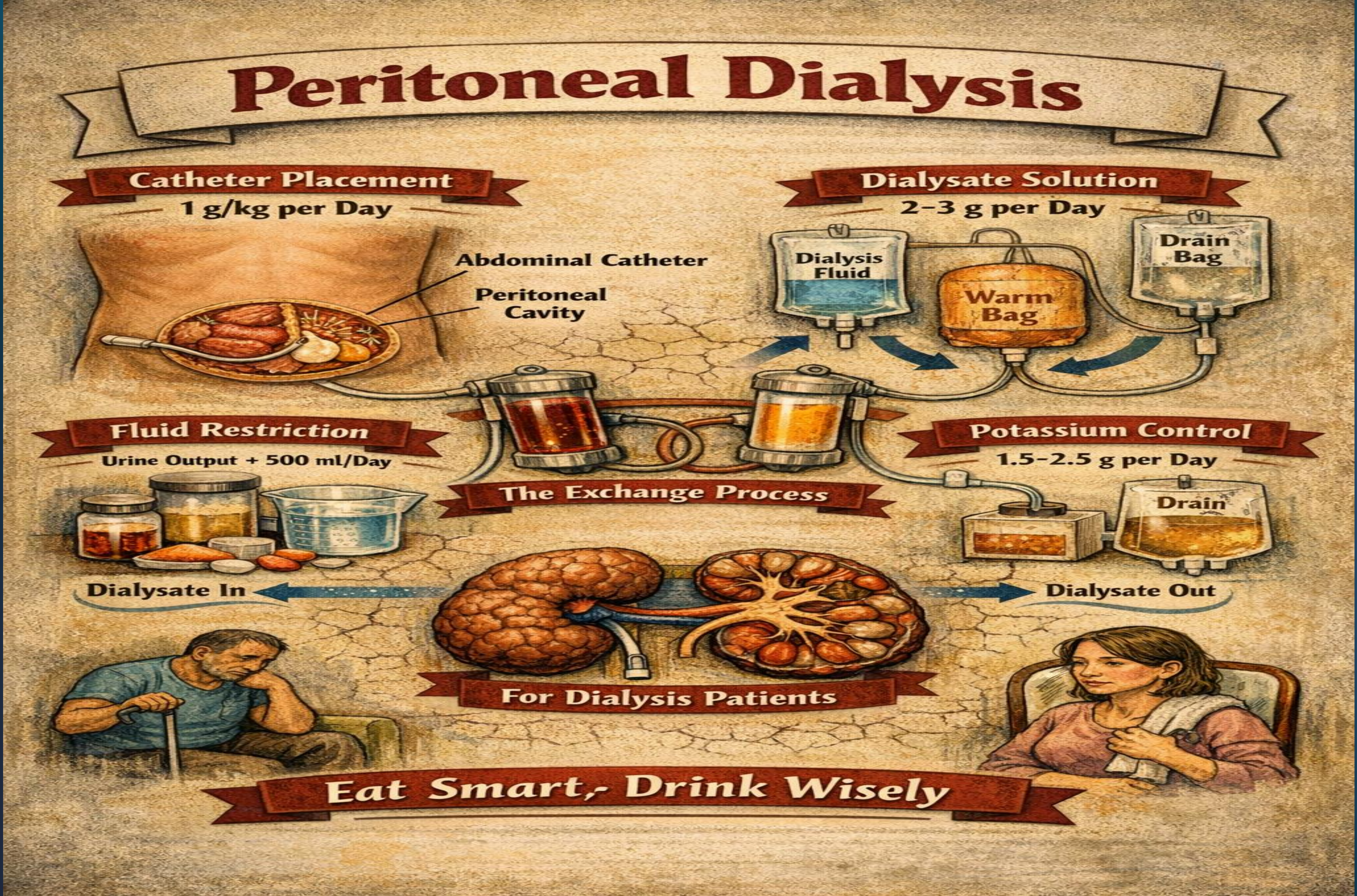

Nutritional & Fluid Therapy
Protein Restriction
1 g/kg per Day
Excess Fluid Volume
Sodium Restriction
2–3 g per Day
CHIPS
Anorexia & Nausea
Fluid Restriction
Urine Output + 500 mL/Day
Potassium Control
1.5–2.5 g per Day
For Dialysis Patients
Fatigue & Weakness
Dependency & Isolation
Fatigue & Weakness
Assess • Educate • Support • Empower
Progression of Kidney Damage & Function Loss

Nursing Management

1. **Meeting psychosocial needs:** The nurse needs to give the patient and family the opportunity to express feelings of anger and concern over the limitations that the disease and treatment impose and over possible financial problems and job insecurity.
   a. Counseling and psychotherapy may be necessary.
   b. The nurse helps the patient to identify safe, effective coping strategies to cope with these problems and fears.

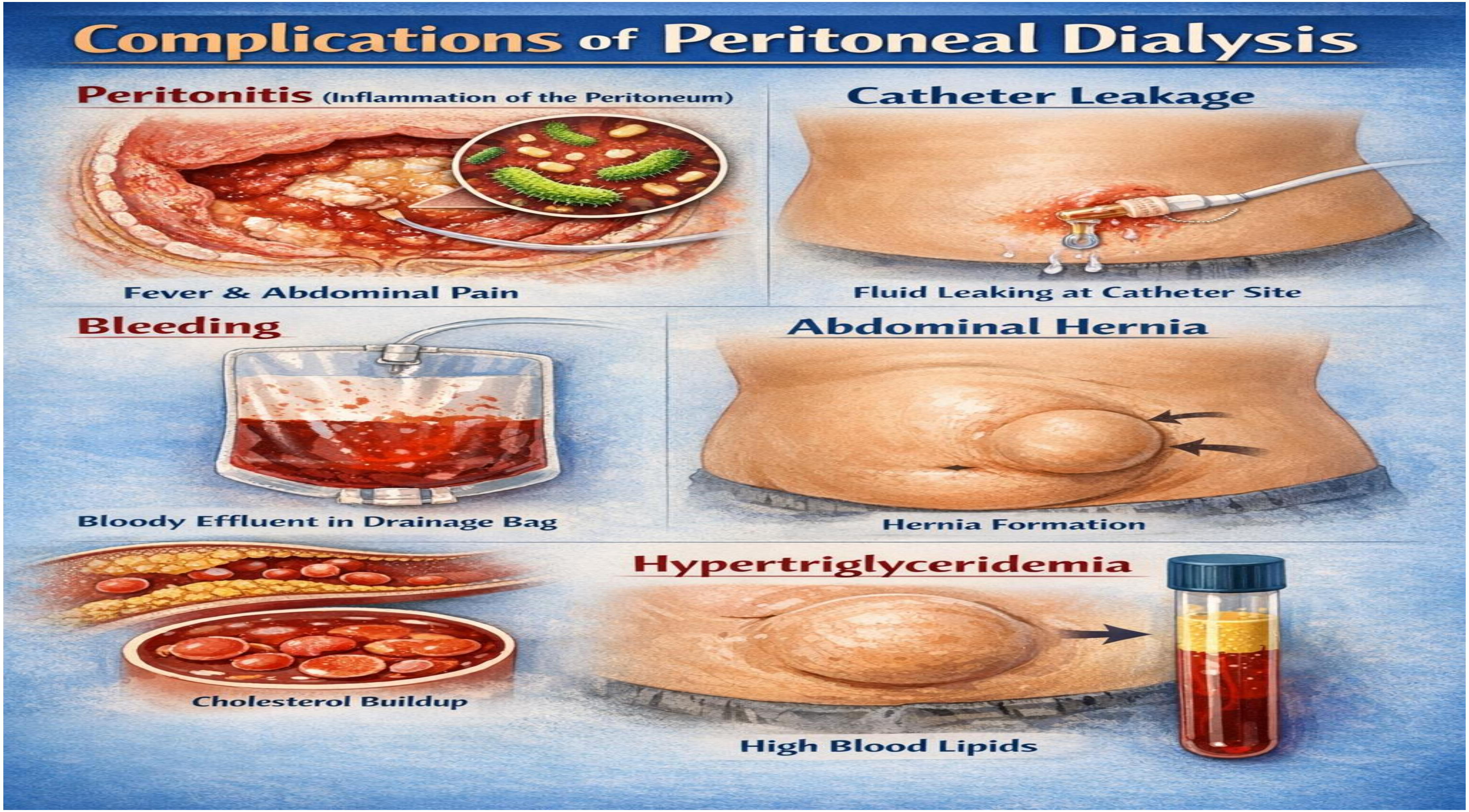
Continuing Care:

The health care team's goal in treating patients with chronic renal failure is to maximize their vocational potential, functional status, and quality of life.


Peritoneal Dialysis
Catheter Placement
1 g/kg per Day
Abdominal Catheter
Peritoneal Cavity
Dialysate Solution
2–3 g per Day
Dialysis Fluid
Warm Bag
Drain Bag
Fluid Restriction
Urine Output + 500 ml/Day
Potassium Control
1.5–2.5 g per Day
The Exchange Process
Drain
Dialysate In
Dialysate Out
For Dialysis Patients
Eat Smart,- Drink Wisely

The goals of peritoneal dialysis are to remove toxic substances and metabolic wastes and to re-establish normal fluid and electrolyte balance. Peritoneal dialysis may be the treatment of choice for patients with renal failure who are unable or unwilling to undergo hemodialysis or renal transplantation.

In peritoneal dialysis, the peritoneum, a serous membrane that covers the abdominal organs and lines the abdominal wall, serves as the semi permeable membrane. Sterile dialysate fluid is introduced into the peritoneal cavity through an abdominal catheter at intervals.



Complications of Peritoneal Dialysis
Peritonitis (Inflammation of the Peritoneum)
Fever & Abdominal Pain
Catheter Leakage
Fluid Leaking at Catheter Site
Bleeding
Bloody Effluent in Drainage Bag
Abdominal Hernia
Hernia Formation
Hypertriglyceridemia
Cholesterol Buildup
High Blood Lipids

Nursing Management

1. *protecting the vascular access*

The nurse assesses the vascular access for patency and takes precautions to ensure that the extremity with the vascular access is not used for measuring blood pressure or for obtaining blood specimens; tight dressings, restraints, or jewelry over the vascular access are to be avoided as well.

Taking Precautions During Intravenous Therapy

When the patient needs intravenous therapy, the rate of administration must be as slow as possible and should be strictly controlled by a volumetric infusion pump.

Monitoring Symptoms of Uremia

As metabolic end products accumulate, uremic symptoms worsen. Patients whose metabolic rate accelerates (those on corticosteroid medications or parenteral nutrition, those with infections) accumulate waste products more quickly and may require daily dialysis. These same patients are more likely to experience complications than other dialysis patients.

Detecting cardiac and respiratory complications

Cardiac and respiratory assessment must be conducted frequently. As fluid builds up, fluid overload, heart failure, and pulmonary edema develop. Crackles in the bases of the lungs may indicate pulmonary edema.

Controlling Electrolyte Levels and Diet

Electrolyte alterations are common, and potassium changes can be life threatening. All intravenous solutions and medications to be administered are evaluated for their electrolyte content. Serum laboratory values are assessed daily

Managing Discomfort and Pain

Complications such as pruritus and pain secondary to neuropath must be managed. Antihistamine agents are commonly used, and analgesic medications may be prescribed.

Monitoring Blood Pressure

Hypertension in renal failure is common. It is usually the result of fluid overload. Many dialysis patients receive some form of antihypertensive therapy and require intense teaching about its purpose and adverse effects.